

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-1972

Brookfield 100 kV
July 25, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1972. The Interconnection Customer proposes a 212.7 MW solar photovoltaic facility interconnecting at the Brookfield 100 kV substation in the Ohio area. The request was received on January 02, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 558 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,928,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Masury Substation - Brookfield Substation 100 KV	Brookfield - Ormond Beach 100 KV L	138.6	125	71.6
Masury - Fondren 100 KV	Brookfield - Ormond Beach 100 KV	135.4	110	71.6
Brookfield - Ormond Beach Substation 100 KV	Brookfield - Masury 100 KV Line	126.1	111	62.4
Ridley Coal - Ormond Each 100 KV	Masury - Brookfield 100 KV Line	113.9	108	62.4
Brookfield 138/100kV Autotransformer	Masry - Brookfield 100 KV line	106.4	68	37.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Masury Substation - Brookfield Substation 100 KV	Rebuild line with higher-capacity conductor	\$1,886,000	100% (\$1,886,000)
Fondren - Masury 100 KV line	Rebuild line with higher-capacity conductor	\$1,238,000	100% (\$1,238,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Brookfield - Ormond Beach Substation 100 KV	Rebuild line with higher-capacity conductor	\$4,574,000	100% (\$4,574,000)
Ridley Coal - Ormond Each 100 KV	Rebuild line with higher-capacity conductor	\$993,000	100% (\$993,000)
Brookfield 138/100kV Autotransformer	Replace transformer with higher-rated unit	\$4,237,000	100% (\$4,237,000)
	Total allocated to project		\$12,928,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.